

FLOCCULENT SPIRAL

The spiral galaxy NGC 2841 is flocculent, with bright stars clumped throughout the disk. Its star formation seems to be caused by local collapses of material rather than a large-scale density wave.

SPIRAL ARMS

The continued presence of spiral arms in most disk-shaped galaxies was once a mystery. If the arms orbit more quickly near the nucleus, then during a galaxy's multi-billion-year lifetime, they would become tightly wrapped around the core. It now seems that the arms are in fact rotating regions of star formation, not rotating chains of stars themselves. The arms arise from a "density wave"—a zone that rotates far more slowly than the galaxy itself. The density wave is like a traffic jam—stars and other material slow down as they move into it and accelerate as they move out, but the jam itself advances only slowly. The increased density helps trigger the collapse of gas clouds and the start of star formation. The strength of the density wave varies between spirals. If the wave is strong, the result is a neat, "grand design" spiral with two clearly defined arms. If it is weak or nonexistent, disk stars will tend to form in localized regions, creating the more clumpy "flocculent" spirals.

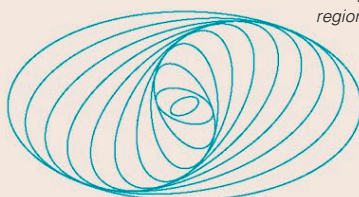
PERFECT GALAXY

In this diagram of an ideal galaxy, objects follow neatly aligned elliptical orbits around the nucleus. They travel fastest when close to the nucleus and slowest when farthest away.



SPIRAL REALITY

In a real galaxy, the orbits do not line up neatly. The variety of alignments, coupled with the slower movement when farther from the nucleus, creates spiral zones in which objects are moving more slowly and so become bunched together.



open clusters of longer-lived stars move out from spiral arm

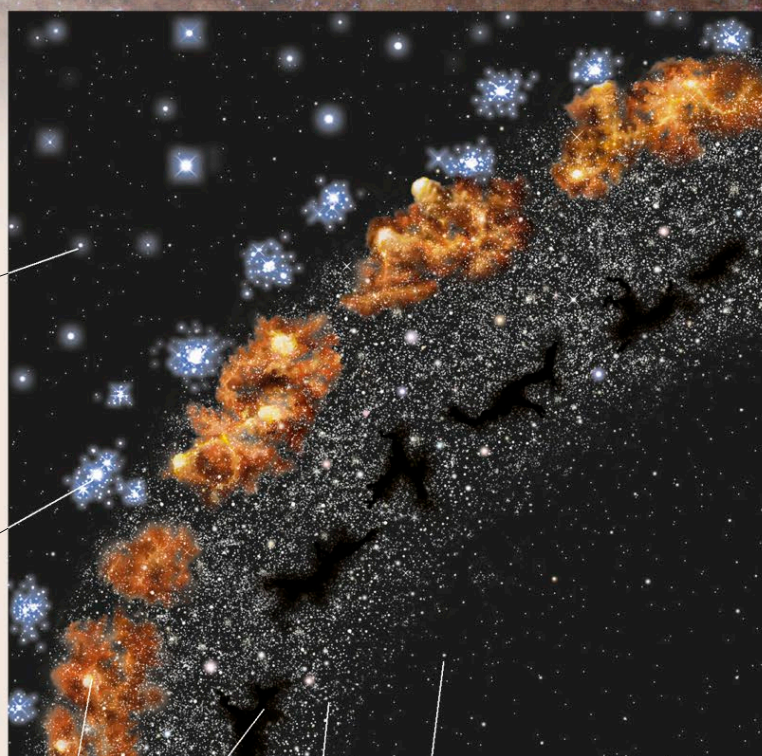
young "OB" star clusters never move far from spiral arm before dying

new stars ignite in HII region (star-forming nebula)

molecular cloud is compressed

density wave causes material to build up

sparse stars orbit faster than the spiral arm and move into arm from behind



DETAIL OF A SPIRAL ARM

As material orbiting in a galaxy's disk approaches the denser region marked by the spiral arm, it packs together. Dark molecular clouds form, some of which turn into star-forming nebulae (see pp.238–239). New stars of all kinds ignite here, but the brightest ones soon die, so they always mark the spiral arms.